# A reading of known relations among the constants: the Standard-Model strong-to-weak ratio at the Planck scale, 1.0768, as a ratio of units, and Koide's lepton relation as a balance between the same and the different

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## Abstract

Run to the Planck mass with the Standard Model at two loops, and with no new particles in between, the measured gauge couplings (PDG 2025, CODATA 2022) give $1/\alpha_3 = 52.98 \pm 0.07$ and $1/\alpha_2 = 49.200 \pm 0.008$ at M_P. Their ratio is $R = 1.0768 \pm 0.0014$ (1.0766 with the formulas of Buttazzo et al. 2013): a Standard-Model number, not a new result. This record proposes one reading of known facts. Inverse couplings are read as counts of readings per elementary cell; the inverse weak coupling as the count in the unit of time, the inverse strong coupling as the count in the unit of space, with counts inversely proportional to the unit. R is then the original ratio between the two units, taken as an initial datum.

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## In brief

Run to the Planck mass with the Standard Model at two loops, and with no new particles in between, the measured gauge couplings (PDG 2025, CODATA 2022) give

$1/\alpha_3(M_P) = 52.98 \pm 0.07$, $1/\alpha_2(M_P) = 49.200 \pm 0.008$, **$R = 1/\alpha_3 \div 1/\alpha_2 = 1.0768 \pm 0.0014$.**

This is a Standard-Model number (Buttazzo et al. 2013 give 1.0766 with the same inputs), not a new result. Everything up to it is standard physics, and so are the structural facts used in this record: the electromagnetic coupling as a mixture of the weak and hypercharge couplings, the dependence of the running on the number of colours and on the particles lighter than the scale, the quadratic weight of charges, and the link between the strong coupling at the Planck scale and the gravitational weakness of the proton (Wilczek).

This record proposes one reading of these known facts. Inverse couplings are read as counts of readings per elementary cell; the inverse weak coupling is the count in the unit of time, the inverse strong coupling the count in the unit of space, and counts are inversely proportional to the unit. R is then the original ratio between the two units, taken as an initial datum. The reading is tested against gravity in three registered forms: as a physical difference between space and time, R would give a post-Newtonian parameter $\gamma = 2R/(1+R)$ or $2/(1+R)$, about 1600 times the Cassini error away (about 59 times the combined error if the working uncertainty of R is included); as a pure ratio of units it gives $\gamma = \beta = 1$. With R and the measured weak coupling, the strong coupling at the Planck scale follows, and through the known running so does $\alpha_s(M_Z)$.

No constant is derived here. Twenty-eight mechanisms for obtaining the strong coupling from the model's own counts were registered before the calculation; none reproduces $\alpha_s(M_Z)$ within two standard deviations, while 0.22 hits would be expected by chance.

A second part (Part B) reads the lepton masses in the same spirit. The amplitudes of the mass term of a family split into a part that the three generations share (the same) and a part that distinguishes them (the different). The proposed reading is that the two weigh the same. This is exactly Koide's relation, which the charged-lepton pole masses satisfy within their measurement uncertainty (about eight parts in a million; the central value lies three parts in a million from 2/3). Applied to neutrinos, the balance has one solution for each mass ordering; with the ΛCDM bound of 0.077 eV (Planck + ACT + DESI DR2, PDG 2025) only the normal-ordering solution remains, with a sum of neutrino masses of $0.0593 \pm 0.0003$ eV (in $w_0w_a$CDM the bound is 0.186 eV and both remain). The same relation was applied to neutrinos by Brannen (2006), with a sum of 0.0600 eV from the data of the time. What the reading adds is a proposed reason, an interpretation, not a new number. For the quarks, three rules registered for how colour shifts the balance do not hold; read in the mirror

(inverse masses), the down-type quarks satisfy the balance within their uncertainties, a relation already in the literature, while the up-type quarks do not.

## How to read this record

The order is: measured data; a direction, declared as a hypothesis; where the numbers meet; and only at the end the reading proposed by the model. For each group of constants that seem to meet, two questions are asked:

1. do these constants meet today, without the hypothesis?
2. are there different hypotheses that make them meet?

Part A (sections 1–8) concerns the gauge couplings: sections 1–4 are standard physics, section 5 is the proposed reading, sections 6–8 say what is known, what was tried and what is not claimed. Part B (sections B and B.2) concerns the lepton and quark masses, in the same order. Every computed number is produced by the code in `code/` and written to `results/`; measured inputs, assigned estimates (such as the 0.02 of section 2) and values quoted from the literature are marked as such.

## 1. The data

All values come from CODATA 2022 and the Review of Particle Physics (sources and fingerprints in `sources/SOURCES.md`; the full table is `results/DATA_TABLE.md`).

| quantity | value | source |
|---|---|---|
| $1/\alpha(0)$ | 137.035999177(21) | CODATA 2022 |
| $1/\alpha^{(5)}$(M_Z), MS-bar | 127.930 ± 0.008 | PDG electroweak review |
| $\sin^2\theta$_W(M_Z), MS-bar | 0.23122 ± 0.00006 | PDG electroweak review |
| α_s(M_Z) | 0.1180 ± 0.0009 | PDG QCD review, world average |
| M_Z, M_W, M_H | 91.1880(20), 80.3692(133), 125.20(11) GeV | PDG summary tables |
| m_t (direct), m_t(m_t) MS-bar | 172.56 ± 0.31, 162.5 (+2.1 −1.5) GeV | PDG summary tables |
| m_e, m_p | 0.51099895069(16), 938.27208943(29) MeV | CODATA 2022 |
| G_N; Planck mass $\sqrt{(\hbar c/G)}$ | $6.67430(15) \times 10^{-11}$ m³ kg⁻¹ s⁻²; $1.220890(14) \times 10^{19}$ GeV | CODATA 2022 |
| γ − 1 (Cassini) | $(2.1 \pm 2.3) \times 10^{-5}$ | PDG, tests of gravitational theory, eq. 21.60 |
| Ġ/G (lunar laser ranging) | $(-5.0 \pm 9.6) \times 10^{-15}$ yr⁻¹ | same review, eq. 21.56 |

## 2. The direction: the couplings seen from the Planck mass

The hypothesis of this section is standard: the three gauge couplings are run from M_Z to the Planck mass with the Standard Model renormalisation-group equations, and no new particle is assumed below M_P. One- and two-loop coefficients are built from the field content and checked against the literature (Machacek and Vaughn); below 172.56 GeV the top quark is removed with a simplified prescription (the left-handed bottom quark counted as half a doublet), which is not a complete electroweak matching; the top Yukawa coupling enters at the threshold. MS-bar scheme, SU(5) normalisation $g_1 = \sqrt{(5/3)}\, g'$ where stated. The end point is the non-reduced Planck mass ($G = \hbar = c = 1$).

| running | $1/\alpha_1$ (SU(5)) | $1/\alpha_2$ | $1/\alpha_3$ | $1/\alpha$_Y (SM) | $R = 1/\alpha_3 \div 1/\alpha_2$ |
|---|---|---|---|---|---|
| **reference: two loops, top threshold** | 33.101 | 49.200 | 52.977 | 55.168 | **1.0768** |

| | | | | | |
|---|---|---|---|---|---|
| one loop | 33.334 | 49.506 | 52.477 | 55.557 | 1.0600 |
| two loops, no top threshold (and y_t(M_Z) = 0.99) | 33.043 | 49.147 | 52.903 | 55.072 | 1.0764 |
| two loops, three-loop QCD term added | 33.101 | 49.200 | 52.976 | 55.168 | 1.0768 |
| end point: reduced Planck mass | 34.159 | 48.394 | 51.173 | 56.932 | 1.0574 |
| end point: 2 M_P | 32.646 | 49.546 | 53.753 | 54.410 | 1.0849 |
| MSSM above 1 TeV (DR-bar) | 17.918 | 23.784 | 28.364 | 29.863 | 1.1926 |

**Cross-check with the literature.** Buttazzo, Degrassi, Giardino, Giudice, Sala, Salvio and Strumia (JHEP 12 (2013) 089, eq. 61) give the Standard Model couplings at the Planck mass with three-loop running and two-loop thresholds; the gauge couplings are given as functions of M_t, α_s(M_Z) and M_W. With the same inputs (α_s = 0.1180, M_t = 172.56 GeV, M_W = 80.3692 GeV) their formulas give $1/\alpha_1 = 33.14$, $1/\alpha_2 = 49.18$, $1/\alpha_3 = 52.94$, hence **R = 1.0766**, against 1.0768 ± 0.0014 here (`code/literature_check.py`). The agreement supports the numbers; it does not show that every omitted term is covered by the estimate below.

**Uncertainty** (`results/uncertainty.log`). On $1/\alpha_3(M_P)$: 0.063 from α_s(M_Z), 0.001 from the three-loop QCD term (a truncation estimate), 0.001 from the top Yukawa coupling (m_t(m_t) ± 1.8 GeV), below 0.001 from the threshold and G_N, and 0.02 assigned to the missing three-loop electroweak terms (an estimate, not computed): **± 0.066** in quadrature. This is a working total of parametric errors and estimated truncation terms, not a measured standard deviation. On $1/\alpha_2(M_P)$: ± 0.008. On R: **± 0.0014**; the parametric inputs alone, propagated directly on the ratio, give ± 0.0013. The value $1/\hat{\alpha}(M_Z) = 127.930$ is quoted by the PDG with α_s = 0.1187 ± 0.0017 and is used here with the world average of α_s, without a joint fit. The MS-bar/DR-bar convention shifts $1/\alpha_3$ by 0.08: this is a choice of definition, not a measurement error.

## 3. Where the numbers meet

| group | question 1: do they meet without the hypothesis? | question 2: other hypotheses that make them meet |
|---|---|---|
| **The three couplings at high energy** | No. In the Standard Model the pairs cross at $1.1 \times 10^{13}$, $2.8 \times 10^{16}$ and $1.7 \times 10^{14}$ GeV. | The MSSM above 1 TeV: the three meet within 0.6 % near $1.3 \times 10^{16}$ GeV, and an exact meeting then requires α_s(M_Z) = 0.1219 (0.1143 for M_SUSY = 10 TeV) against 0.1180 ± 0.0009. Also SO(10) with an intermediate scale, and extra dimensions (Dienes, Dudas, Gherghetta): not computed here. |
| **The band 49–55 of the three couplings at M_P** | No: the band is 12 % wide (60 % with the SU(5) normalisation). | Any number in the band. Among 736 "simple numbers" between 40 and 60, 79 lie within 2.3 % of the weak coupling and 9 within twice the working uncertainty of the strong one (`results/meetings.log`). |
| **Layers ln(M_P/m) against the couplings at M_P** | The electron layers, 51.53, lie 1.45 below the strong coupling: 22 times its working uncertainty. | Chance: 2 layer-coupling pairs within 3 % out of 64, 2.1 expected. |
| **The proton's gravitational weakness** | Yes, by known physics: given m_p/Λ from QCD, $\alpha_G(p) = (m_p/M_P)^2 = 5.9 \times 10^{-39}$ is explained by the running of the strong coupling (Wilczek); this record uses the measured m_p and does not predict it. A shift of 0.07 in $1/\alpha_3(M_P)$ changes α_G(p) by a factor 0.91 at fixed m_p/Λ. | Not needed. |
| **Large numbers (Dirac, Eddington)** | $e^2/(G\, m_e\, m_p) = 2.27 \times 10^{39}$ is arithmetic of α and the two layers. | Dirac's $G \propto 1/t$ needs $\dot{G}/G$ of order $10^{-10}$ per year: excluded by lunar laser ranging by a factor of about 7000. Eddington's integer 137 differs from 1/α(0) = 137.035999177(21) by 0.036, about 1.7 million times the measurement uncertainty. |

The numbers meet only where known physics makes them meet. The historical case, the three couplings and supersymmetry, shows the logic of the two questions: three constants, a meeting that costs two parameters (the scale and the common value), one prediction left (α_s), which the measurement puts under tension.

## 4. What R depends on

R is defined for the Standard Model with no new particles below M_P, at two loops, in MS-bar, with the non-reduced Planck mass as end point. With one loop it is 1.0600; with the reduced Planck mass 1.0574; with 2 M_P 1.0849; with the MSSM it is not defined in the same way (1.1926 at M_P). The choice of the end point is part of the reading: the model takes the Planck scale as the scale where gravity has one reading per cell, which is the non-reduced Planck mass.

## 5. The reading proposed by the model

This section uses the language of the model; the picture behind it is described in `WHERE_THIS_COMES_FROM.md`, and nothing in sections 1–4 depends on it.

**5.1 Counts.** In the model the state of a place is a register of T readings over K categories, updated by Bayes' rule (a Dirichlet posterior). The coupling of a cell is read as the inverse of its number of readings. The information distance covered by one new reading, measured with the local quadratic form of the Fisher metric, has expected square

$$E[\Delta s^2] = (K - 1)/(T + 1)^2,$$

for every composition of the register (checked symbolically for K = 4, `results/reading.log`; T includes any prior pseudo-counts). The finite Fisher–Rao distance agrees with it only when every category holds many readings: for a uniform register with T = 4 its exact expectation is 13 % smaller, at T = 1000 the difference is 0.1 %; near the edge of the simplex many readings in total are not enough (counts (1, 1, 1, 999997), T = $10^6$: 31 % smaller). The root-mean-square step per reading therefore goes as $\sqrt{(K - 1)/(T + 1)}$: **at fixed K, T + 1 is inversely proportional to the unit step.**

**5.2 Time, space and the two couplings.** The reading assigns the weak coupling to the unit of time and the strong coupling to the unit of space. Holding fixed the ratio between the cell and the unit of time, expressing time in the unit of space divides the cell by the original ratio between the two units and, by 5.1, multiplies T + 1 by the same ratio; the reading identifies the count $1/\alpha$ with T, a convention discussed below. The data then fix that ratio:

**R = 1.0768 ± 0.0014:** at the origin, the count in the unit of space is 7.7 % larger than in the unit of time, that is, with the step taken as 1/T (the convention of this reading), the unit step of space is 7.1 % smaller.

In the model this ratio is an initial datum: it is not derived and needs no further justification. The identification uses the counts $1/\alpha$; with T + 1 in place of T the ratio of the steps would be 1.0753, a difference of 0.0015, about the size of the uncertainty of R: the convention is part of the reading.

**5.3 The test with gravity.** If R is the original ratio between space and time, it must appear wherever a measure of space is compared with a measure of time. In the model's gravitational sector the lapse N (time) and the conformal factor $\psi$ (space) are tied by $\psi + N\psi = 2$, which gives the exterior Schwarzschild metric in isotropic coordinates. Three ways of inserting R (registered before the calculation, `records/REGISTRAZIONE_R_ALTROVE.md`):

| how R enters | $\gamma$ | $\beta$ | against Cassini |
|---|---|---|---|
| R weighs the time part: $\psi + R\,N\psi = 1 + R$ | $2R/(1+R) = 1.0370$ | $(3R+1)/(2(1+R)) = 1.0185$ | 1607 × the Cassini error |
| R weighs the space part: $R\psi + N\psi = 1 + R$ | $2/(1+R) = 0.9630$ | $(R+3)/(2(1+R)) = 0.9815$ | 1608 × the Cassini error |
| R is a ratio of units only | 1 | 1 | compatible |

Including the working uncertainty of R (linear propagation), the first two forms are about 59 times the combined error away. Of the three forms tested, only the third is compatible with gravity: R as a ratio of units, invisible to gravity and visible in the couplings because readings are counted per cell, that is, in the unit. This does not show that every other physical reading of R is excluded, and the third form is not a new gravitational prediction: it keeps the relation that already gives $\gamma = \beta = 1$.

**5.4 What R does not fix.** The hypercharge coupling (the part of electromagnetism that the reading calls the descriptor) is not fixed by R with any of eight simple rules: the descriptor reading in the unit of time, of space, or of both (sum or mean), with the Standard Model or the SU(5) charge unit. Closest miss: 1.3 in $1/\alpha_1$. It remains a separate datum.

**5.5 What follows.** With R and the measured weak coupling, the strong coupling at the Planck mass follows, and with the

known running so does α_s(M_Z). The proton's gravitational weakness $10^{-39}$ is linked to it only through the QCD relation between the scale Λ and the proton mass, taken from standard physics: the value in `results/data.log` is computed from the measured m_p and M_P, and this record does not predict the proton mass. In the reading, the $10^{-39}$ is the ratio of the two units at the origin carried down by the running of the strong coupling. The number of inputs is the same as in the Standard Model: one datum (R) in place of one measured constant (α_s).

## 6. What is known and what is proposed

| correlation used | in standard physics? |
|---|---|
| electromagnetism as a mixture of the weak and hypercharge couplings | yes: Glashow 1961, Weinberg 1967, Salam 1968 |
| the strong coupling runs faster than the weak one because of its colours | yes: Gross and Wilczek, Politzer, 1973 |
| the three couplings nearly meet at high energy | yes: Georgi, Quinn, Weinberg, 1974 |
| heavy particles decouple from the running below their mass (in MS-bar this is implemented with thresholds and matching) | yes: Appelquist and Carazzone, 1975 |
| the weight of a charge is quadratic | yes: Born's rule and quantum electrodynamics |
| the proton's $10^{-39}$ comes from the running of the strong coupling | yes: Wilczek, 2001 |
| the electron has three times the charge of the d quark | yes, from anomaly cancellation (Geng and Marshak; Babu and Mohapatra; Minahan, Ramond and Warner, 1989–90) |
| a ratio of space and time units is invisible to gravity | yes: c as a conversion factor |
| couplings as counts of readings per cell | proposed here |
| the strong-to-weak ratio at the origin as the ratio of the units of space and time | proposed here, as far as the author knows |
| different couplings for the space and time directions | known in another sense: on anisotropic lattices the two couplings belong to the directions of one gauge theory (Karsch 1982; Carena et al. 2022), not to two different forces |
| the step per reading, inversely proportional to the readings, as the unit of time | the information geometry is known (Fisher, Amari); its use as the unit of time is proposed here |

Standard physics gives each correlation its own reason: the symmetry group, the charges, the running. The reading proposes a common interpretation: the same original cell counted in the unit of time, in the unit of space, or by the descriptor. It does not derive the gauge groups, the field content or the charges, which enter through the running. The numbers coincide with those of standard physics, as they must if both describe the same world.

## 7. Registered attempts to obtain the strong coupling from the model's counts

Before the calculation, 28 mechanisms were written down (`records/REGISTRAZIONE_FORTE_MM.md`, SHA-256 37c3ac7b…, time recorded locally 27 Sep 2026 11:40:19). They come in four families generated by fixed rules: the entropy A/4 of every surface of the model's minimal saturated place, in nats and in bits (12); the exponent between three dimensions and the two-dimensional mirror of the model (4); the model's count 16π placed at scales near M_P, with the running in between (6); 16π plus a small count of the model (6). Each was carried to α_s(M_Z) with the reference running.

Result: **0 hits out of 28** (tolerance ± 0.0018 on α_s(M_Z); 0.22 hits expected by chance, a probability of 20 % of at least one). The closest are 16π + 3 (α_s = 0.1140) and 16π + 2 (0.1292). For eight candidates ($1/\alpha_3$ below 34) there is no solution for α_s(M_Z) up to 0.9, the largest value tried; for five (A9–A12 and B2, $1/\alpha_3$ between 85 and 356) no solution was found in the range explored with these equations: the running reaches M_P only for α_s(M_Z) above about 0.0264, and at that edge the top Yukawa coupling at M_P is about 6000, far outside the perturbative regime

(`results/strong_from_counts.log`). All are misses. The model's count 16π at M_P gives α_s(M_Z) = 0.1765; the corresponding proton gravitational coupling, about 40 times larger than measured, is only indicative, because the recipe for Λ is then used beyond its range ($\Lambda^{(3)}$ above m_c).

Two expressions noticed before the registration, 16π + e and 16π at M_P/(4π), reproduce α_s(M_Z) (0.1179 and 0.1163). They were found by looking at the target and are reported as examples of how easily a known number is matched after the fact.

## 8. What is not claimed

- No constant is derived. R is read from the data; its role as the original ratio of units is the proposed reading.
- R depends on the choices of section 4.
- In standard physics couplings are dimensionless and do not depend on units. The reading assumes that readings per cell are counted in the unit of the corresponding measure. If the two forces saw different ratios of space and time in their kinematics, particles would have different limiting speeds; Coleman and Glashow give $1 - c^2 < 10^{-23}$ (cosmic-ray protons, one-sided) and $|1 - c^2| < 6 \times 10^{-22}$ (spectroscopy), with c the speed of light in units of the maximal attainable speed of matter; the bounds are much weaker for other particles, and none of them bounds R directly. The reading places R in the counting, not in the motion, which is the same conclusion reached with gravity in 5.3.
- 16π, a working value used in the model for the count of the original cell, is not the count of any coupling at M_P: it lies 1.07 from the weak coupling (135 times its uncertainty) and 2.71 from the strong one (41 times its working uncertainty).
- The hypercharge coupling is not fixed by R.

## B. The lepton masses: a balance between the same and the different

**Data** (`results/DATA_TABLE.md`). Charged-lepton masses (CODATA 2022; m_τ = 1776.93 ± 0.09 MeV, PDG). Neutrino mass-squared differences (PDG 2025 lepton summary): $\Delta m^2_{21} = (7.50 \pm 0.19) \times 10^{-5}$ eV², $\Delta m^2_{32} = (2.451 \pm 0.026) \times 10^{-3}$ eV² (normal order) or $(-2.527 \pm 0.034) \times 10^{-3}$ eV² (inverted order). Upper bounds on the sum of neutrino masses at 95 % CL (PDG 2025, Neutrinos in cosmology, Table 26.2): 0.077 eV (Planck + ACT + DESI DR2), 0.048 eV (with SPT-3G), 0.053 eV (PR4 + ACT + DESI DR1 + Lyman-α).

**Direction (a hypothesis).** In a family of three particles with the same charges, the mass term couples the left and right components of the three generations: a 3 × 3 matrix whose singular values are the masses. Let A be a Hermitian matrix of amplitudes whose eigenvalues a_i satisfy $a_i^2 = m_i$: in the mass basis $A = \mathrm{diag}(a_1, a_2, a_3)$, and an amplitude can carry a sign. This is a construction in the mass basis, not a property of every chiral mass matrix. A splits in one way only into a part shared by the three generations and a traceless part: $A = (\mathrm{Tr}\, A/3)\, I + A_0$. For every Hermitian A with $\mathrm{Tr}\, A \neq 0$, with the Hilbert–Schmidt norm (checked symbolically, and on 1000 random real symmetric and 1000 complex Hermitian matrices, `results/masses.log`):

$$Q = (m_1 + m_2 + m_3)/(a_1 + a_2 + a_3)^2 = (1/3)\,(1 + |A_0|^2/|A_{id}|^2).$$

**Where the numbers meet.**

| family | result |
|---|---|
| charged leptons | Q = 0.6666645 ± 0.0000051, so that the ratio of the different to the same is 0.999997. The balance requires m_τ = 1776.97 MeV (measured 1776.93 ± 0.09). |
| neutrinos, normal order | one solution, with the negative amplitude on the lightest state: m = 0.00036, 0.00867, 0.05026 eV; sum 0.0593 ± 0.0003 eV |
| neutrinos, inverted order | one solution, negative amplitude on the lightest state: sum 0.1029 ± 0.0007 eV |
| quarks (section B.2) | Q = 0.85 (u, c, t), 0.73 (d, s, b) with the PDG masses |

With all amplitudes positive there is no solution for neutrinos; the negative sign is not chosen: it is the only case with a solution, and it falls on the lightest state in both orderings. The oscillation minima of the sum are 0.0589 eV (normal) and 0.0998 eV (inverted). The ΛCDM bound 0.077 eV allows the normal-order solution and excludes the inverted one; the bounds depend on the cosmological model, and the $w_0w_a$CDM bound in the same table (0.186 eV) allows both. The two tighter bounds (0.048 and 0.053 eV) exclude every normal-order spectrum allowed by oscillations, including its

minimum: they express a tension between cosmology and neutrino masses as such, not a test of this reading.

- Question 1: the balance is today a known empirical relation (Koide) that holds with the pole masses to one part in $10^5$ and with the running masses only to one part in $10^3$ (Sumino, JHEP 05 (2009) 075). Its geometric form, a 45° angle between the vector of the √m_i and (1, 1, 1), was given by Foot (1994); the split into the same and the different is a rewriting of it, not a new identity of the masses.
- Question 2: Brannen (2006) applied it to neutrinos with a negative root for the lightest state and found m = 0.000388, 0.00895, 0.0507 eV (sum 0.0600 eV); the numbers above are the same relation with 2025 data (sum 0.0593 eV), not identical numbers. Sumino (2009) proposed a family gauge symmetry that explains why the relation holds with pole masses.

**The reading proposed by the model.** The matrix of a family imports «the same» and «the different» in equal measure: $|A_0| = |A_{id}|$. What is imported and not derivable is the scale of the family and the orientation of the different part (two of the three masses); what is structural is their balance (the third). The reading was stated for families without colour; this restriction was written knowing that the quarks do not satisfy the relation. For neutrinos the mechanism that gives them mass is not specified: the calculation uses only the three mass eigenvalues. In the model, mass is the frequency of the internal clock of a particle, which is the pole mass, and the relation holds precisely for pole masses. The registration of this part (`records/RIFINITURA_E_REGISTRAZIONE.md`, SHA-256 ec65f7c6…, local time 27 Sep 2026 14:08:45) declares what was known before the calculation.

**What is not claimed.** No new number: the relation is Koide's and the neutrino prediction is Brannen's, updated. The reading proposes a reason for a known relation; the equal weight of the same and the different remains an interpretive choice, not a dynamical explanation. Individual masses are not derived. Attempts made during the same work on the Higgs self-coupling and on the top Yukawa coupling were not blind and are not reported as results.

## B.2 The quarks: the balance in the mirror

**Data** (`results/DATA_TABLE.md`). Quark masses from the PDG 2025 summary table: u, d, s in MS-bar at 2 GeV (2.16 ± 0.07, 4.70 ± 0.07, 93.5 ± 0.8 MeV); c and b in MS-bar at their own mass (1.2730 ± 0.0046, 4.183 ± 0.007 GeV), all five with uncertainties at 90 % CL; t from direct measurements (172.56 ± 0.31 GeV) or in MS-bar (162.5 +2.1 −1.5 GeV, no confidence level stated, taken here as ± 1.8 GeV at one standard deviation). Quarks are never observed alone and the light ones have no pole mass, so the balance of a quark family depends on how the masses are defined. The masses are therefore also carried to one common scale, M_Z, with the QCD running of α_s and of the masses (`code/quarks.py`).

**Where the numbers meet** (`results/quarks.log`).

| family | Q, PDG masses (t direct) | Q, PDG masses (t MS-bar) | Q, all masses at M_Z |
|---|---|---|---|
| u, c, t | 0.8488 | 0.8449 | 0.8868 |
| d, s, b | 0.7313 | 0.7313 | 0.7470 |

The definition of the masses moves Q by up to 0.042; for the charged leptons the uncertainty on Q is 0.000005.

**Three rules registered for how colour shifts the balance** (`records/REGISTRAZIONE_QUARK.md`). Write $\rho = |A_0|/|A_{id}|$ for the ratio of the different to the same, so that $Q = (1 + \rho^2)/3$; for the leptons $\rho = 1$. Three rules were written from the premises of the reading before the calculation: (1) the different part of a quark family is read in the unit of space and the same part in the unit of time: $\rho = R$ for both families; (2) colour has three directions, and the model's comparison in three dimensions carries the power 3/2: $\rho = R^{(3/2)}$ for both; (3) the weight grows with the readings shared, that is with the charge in thirds (2 for the up family, 1 for the down family): $\rho = R^2$ and $\rho = R$. A rule passes if it holds for both families at M_Z, within a tolerance equal to the spread between the definitions plus twice the PDG uncertainties (0.043 up, 0.017 down).

| rule | Q predicted, up / down | up family (0.8868) | down family (0.7470) |
|---|---|---|---|
| $\rho = R$ | 0.7198 / 0.7198 | no | no |
| $\rho = R^{(3/2)}$ | 0.7495 / 0.7495 | no | yes |

| | | | |
|---|---|---|---|
| ρ = R² (up), R (down) | 0.7814 / 0.7198 | no | no |

None passes. The single hit is what chance allows: with tolerances this wide, a value of Q taken at random between 0.6 and 0.95 falls within the tolerance of at least one of the six rules $\rho = R^n$ (n = 1/2, 1, …, 3) in 71 % (up) and 55 % (down) of cases; the expected numbers of rules within tolerance are 1.48 and 0.59. The data would require n = 3.43 (up) and 1.46 (down). Before the calculation it had been noticed, and declared, that ρ of the up family is close to $R^3$; with no logic written in advance, that rule was not tested. The value $(1 + R^2)/3 \approx 0.720$ had also been noticed near the down family in the registration of Part B, before rule (1) was written: the rules are not independent of the known values.

Two departures from the registration were found afterwards and are declared (`results/quarks.log`, part 7): the registration states two loops for the running of the masses, and the calculation uses three; it states two standard deviations of the masses, while the quoted uncertainties of u, d, s, c and b are at 90 % CL, so that for them the tolerance used corresponds to about 3.3 standard deviations. Redone as registered (two loops; two standard deviations for u, d, s, c, b, the top kept at ± 1.8 GeV), Q at M_Z becomes 0.8854 (up) and 0.7463 (down) and the tolerances 0.042 and 0.016; adding the uncertainty of α_s in the running of the masses they stay 0.042 and 0.016 (0.044 and 0.017 with three loops). No outcome changes. The spread between definitions is a working tolerance, not a statistical interval.

**The mirror** (`records/REGISTRAZIONE_QUARK_SPECCHIO.md`). For two numbers, greatest common divisor × least common multiple = product: on a logarithmic scale the common multiple is the mirror image of the common divisor, with the mirror at the geometric mean. A mass read in the unit of the common multiple therefore goes as the inverse mass. Two predictions were registered: Q(1/m) = 2/3 for both quark families; and, reading a family as m^p, the same exponent p for both families.

| family | Q(1/m), PDG masses | Q(1/m), at M_Z | p < 0 with Q(m^p) = 2/3 (PDG / M_Z) | p > 0 (PDG / M_Z) |
|---|---|---|---|---|
| u, c, t | 0.9178 | 0.9140 | −0.514 / −0.512 | +0.610 / +0.550 |
| d, s, b | 0.6647 | 0.6674 | −1.005 / −0.998 | +0.859 / +0.829 |
| e, μ, τ (pole masses) | 0.8514 | – | −0.651 | +1.000 |

As registered, neither prediction is satisfied: the up family does not balance in the mirror, and the two families require different exponents. The down family does balance in the mirror:

- moving d, s and b to the ends of their PDG intervals (90 % CL, without correlations), Q(1/m) ranges from 0.6623 to 0.6670, and 2/3 lies inside; this is not a 90 % interval on Q;
- exactly 2/3 requires m_s/m_d = 20.28, against 19.89 from the central values; the PDG summary table gives m_s/m_d = 17–22;
- all three masses take part: without the bottom quark the same expression gives 0.70;
- the exponent required is −1.005 (PDG masses) and −0.998 (at M_Z), compatible with −1, the pure mirror.

A third registered check, weighting the three up quarks with 1, R and 2R/(1+R) (the value of the cell in time, in space, and in the part that mirrors itself), moves Q by less than 0.01, as expected: it does not reach 2/3.

- Question 1 and question 2: the mirror balance of the down-type quarks is already in the literature. Rivero (2026) proposes exactly this relation, $(1/m_d + 1/m_s + 1/m_b)/(1/\sqrt{m_d} + 1/\sqrt{m_s} + 1/\sqrt{m_b})^2 = 2/3$, finds 0.66750 at M_Z (0.6674 here) and exactly 2/3 near 280 TeV under Standard-Model running (a scale that depends on the inputs and on the implementation of the running; the code of this record runs the masses only up to M_Z). The mixed triplets are also known: (c, b, t) close to 2/3 (Rodejohann and Zhang 2011; Kartavtsev 2011; Cao 2012, who finds 0.6688), and (s, c, b) with a negative root for s (Rivero 2011). With the PDG masses they give 0.669 and 0.675 here; with all masses at M_Z, 0.720 and 0.697: they come close to 2/3 with masses defined at different scales and move away at the common scale M_Z tried here.

**What is proposed and what is not.** The picture that comes out is: the charged leptons balance with their direct masses, the down-type quarks in the mirror, the up-type quarks in neither. For the down-type quarks the reading proposes a reason, the mirror between the common divisor and the common multiple (the step from integers to the inverse of real masses is a hypothesis of the reading), as it proposes one for the leptons; it gives none for the up-type quarks, and it does not say why the leptons are read directly and the down-type quarks in the mirror. No rule linking leptons and quarks was found. The relation of the down-type quarks is not new.

## 9. Reproduction

Python 3.10 or later with numpy, scipy and sympy. From the folder `code/`:

python run_all.py

The script runs `data.py`, `running.py`, `literature_check.py`, `uncertainty.py`, `meetings.py`, `strong_from_counts.py`, `reading.py`, `masses.py` and `quarks.py` in this order, writes their logs and JSON files in `results/`, and lists the SHA-256 fingerprints of code and results in `results/SHA256SUMS_run.txt`. A few minutes on a four-core desktop.

## 10. Records

`records/` holds the five registrations written before the calculations of sections 5.3, 7, B and B.2, in Italian, with their fingerprints and the local times (`SHA256SUMS_REGISTRAZIONE.txt`, `SHA256SUMS_REGISTRAZIONE_MASSE.txt`, `SHA256SUMS_REGISTRAZIONE_QUARK.txt`). The times are recorded by the author's machine; they are not an external timestamp. `records/README.md` summarises them in English.

## 11. Glossary

- **Reading, readings per cell.** In the model, the elementary event by which a place registers something; the number of readings in the elementary cell is read as the inverse of a coupling.
- **Unit of time, unit of space.** The elementary step of the cell measured as time or as space.
- **Descriptor.** In the model, the part of the description that makes an interaction electromagnetic; here it corresponds to the hypercharge coupling.
- **Layers.** ln(M_P/m): the number of e-folds of scale between the Planck mass and a particle mass.
- **Minimal saturated place.** In the model, the smallest place whose count is saturated; its entropy A/4 is 16π in the model's units.

## 12. References


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## Author


The author is an independent researcher, not a physicist. The calculations and the text were produced with AI assistants; every computed number is produced by the code of this record. Registrations were written before the calculations they concern, as stated in section 10.